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"It's Just [Complicated] Sleep": Discourses of Sleep and Aging in the Media

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Abstract

The media are influential in shaping beliefs and attitudes on aging and health-related behaviors. Sleep is increasingly recognized as a key pillar for healthy aging. However, the role of media representations of sleep is yet to be assessed with regard to discourses of aging. Texts from New Zealand's main free online news source were collated using key words "sleep" together with "aging," "older," "elderly," or "dementia" between 2018 and 2021. Contents of 38 articles were interpreted using critical discourse analysis. Discursive constructions described an inevitable decline of sleep with aging, including impacts of both physiological decline and life stage transitions; sleep's role as both a remedy and risk for ill health and disease; and the simplification of solutions for self-managing sleep juxtaposed alongside recognition of its complexity. The audience of these complex messages is left in the invidious position of both pursuing sleep practices to prevent age-related decline, whilst also being told that sleep degradation is inevitable. This research demonstrates the complexity of media messaging and the fraught options it offers: good sleep as both a reasonable achievement to strive for and as impossibly idealistic. Findings mirror two predominant health identities available to older people, as responsible for resisting aging or as falling into inevitable decline. This reveals additional expectations around appropriate time use and behaviors with aging. More nuanced messaging that goes beyond sleep as a resource for health and waking productivity is recommended. Acknowledging the complexity of sleep, aging, and society could be the starting point of such adaptation.

Keywords: Critical discourse analysis, Dementia, Insomnia, News, New Zealand

Representations of aging circulate through social science theory, research and policy, and are reflected and reproduced through media and popular culture representations of what it means to be an older person (Laliberte Rudman, 2006). Representations of older age are not homogeneous, they work to separate those who are merely older from those who are too old (Gilleard & Higgs, 2016). Active and productive older people are differentiated from vulnerable, passive, and dependent older people (O'Neill & Ní Léime, 2022; Reul et al., 2022). Swacha (2017) found two dominant metaphors for aging contained in public health discourse: 'aging as pathology' where aging is viewed as undesirable and something to be cured; and "aging is ageless," where people can live as if they are not aging by demonstrating levels of activity, health, and productivity associated with younger people (Swacha, 2017). These possibilities are not neutral; older people who are healthy and active are regarded as aging successfully and those who are frail and ill are deemed personally responsible (Rozanova, 2010).

The representations of healthy aging in later life compare active and engaged older people with frail and passive older people who function as an explicit archetype of what not to become. Harris et al. (2016) examined a health promotion video that explicitly drew on this comparison by using a split screen to display these two versions of aging. A healthy older man was presented alongside a man with chronic illness and the audience was invited to decide which aging future they will have. Audiences responded variously, both applauding the message that health was a personal choice and resisting both the individual responsibility to remain healthy and the implication that living with chronic illnesses meant a life not worth living. Such representations of individual responsibility for aging well are ubiquitous in public health communications and highlight personal responsibility for successful aging (Rozanova, 2010), as well as the responsibility to manage or avoid age-related conditions such as dementia (Petersen & Schicktanz, 2021), vision loss (Fuchigami et al, 2022), and social isolation (Breheny & Severinsen, 2018). This representation of age-related decline as a personal responsibility is a powerful message that older people receive and take up as a moral responsibility for self-care (Clarke & Bennett, 2013).

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Media Representations of Aging

The media are influential in shaping beliefs about aging and attitudes to older age (Edström, 2018; Makita et al., 2021; Vasil & Wass, 1993). Older people and later life are not routinely considered newsworthy; research has noted that there is a deficit of news media content focusing on older people (Edström, 2018; Iversen & Wilińska, 2020; Kovács et al., 2021). When older people are included, media representations tend to associate older age with negative characteristics, including incompetence and declining health (Fealy et al, 2012; Iversen & Wilińska, 2020). Popular media headlines often encourage older people to resist aging, suggesting that aging is a choice (Breheny & Stephens, 2019). New Zealand media typical of this style of reporting includes the headline: "Exercise more and live longer, elderly urged," followed by accompanying text focusing on the benefit this will bring to "the already stretched public health system." This illustrates the two predominant health identities available to older people, as personally responsible for resisting aging or as declining into dependence. Separating the fit from the frail requires the media to use a variety of strategies and markers of older age. This includes appearances and activities that align with everyday criteria for marking out people as "old" (Iversen & Wilińska, 2020). One area receiving increasing attention is the attainment of sufficient sleep and the associated practices surrounding sleep indicative of optimizing waking health.

Sociology of Sleep

Although the science of sleep is conventionally grounded in chronobiology, physiology, and medicine (Assefa et al., 2015; Dement, 2008), an emerging body of research recognizes the important role of society (Grandner, 2019; Meadows, 2005; Williams, 2005). Sleep is a practice requiring negotiation between the self, others, and wider society. Socio-ecological models of sleep illustrate how beliefs and culture, living environment, family, and wider society play a role in sleep-related expectations and practices (Grandner, 2019; Meadows, 2005; Robbins, Jean-Louis, et al., 2019).

The architecture and structure of sleep as well as sleep-related behaviors change over the life course. Periods of significant life transition such as retirement and care responsibilities typically influence sleep regulation and quality (Gibson et al., 2023; Myllyntausta et al., 2021). Prevalence of insomnia, "problem sleep," and sleep disordered breathing have been reported to increase with age. A rise in sleep disorders has been associated with neurophysiological changes with advancing age, changing exposures to external time cues (such as regularly timed bright light, meals, and exercise), health conditions and their associated medications, as well as the individual's psychosocial context and sleep related practices (commonly referred to as "sleep hygiene"; Reid et al., 2006; Roepke & Ancoli-Israel, 2010). Such factors have been found to be particularly volatile among people with dementia, contributing to unique and more severe sleep disruptions (Gibson et al., 2014). Those with an illness or sleep disorder invariably disrupt those who they share a bed or bedroom with. This can be challenging in informal care situations with regard to care provision, functionality while awake, and overall well-being of household members (Gibson et al., 2023).

Beliefs around sleep needs, behaviors, and help-seeking vary across the life course, which is important clinically, socially, and academically (Staton et al., 2019). A recent interview study with older New Zealanders found they commonly made comparisons between their own sleep and a socially perceived "ideal" sleep (Crestani et al., 2022). Sleep transitions were presented as a gradual and accepted part of aging, and so participants typically resisted seeking medical help or treatments for sleep disruptions. Although they were aware of ways to enhance their sleep, many did not prioritize practices to improve sleep. The complexity of older people's responses indicates a need to explore discourses of sleep to better understand sleep health messaging for older people.

Media Representations of Aging and Sleep

The media are powerful in terms of shaping both what issues are newsworthy and how people think about those issues (Rozanova et al., 2016). Media are crucial in the circulation of ideas regarding optimal aging and sleep. They influence the way people interpret sleep practices and problems, where responsibility lies, and potential solutions (Kitzinger, 2007). The rapidly growing field of sleep science has been met with greater public interest and therefore increased media representation of sleep-related health messaging. Previous research has described the medicalization of sleep problems and treatments (Kroll-Smith, 2003; Seale et al., 2007; Varallo et al., 2022; Williams et al., 2008a, 2008b), gendered constructs of sleep (Zarhin, 2021), and the malleability of sleep around working life (Boden et al., 2008; Williams & Boden, 2004). These studies indicate that sleep is typically described using medical perspectives and by sensationalizing "harmful" behaviors. Messages, therefore, are often divorced from both scientific rigor and the realities of individual and social circumstances (Barbee et al., 2018; Williams & Boden, 2004). Content can also entrench broader social discourses and stigmas relating to sleep disorders, sleepiness, and disease (Robbins, Grandner, et al., 2019). Such research indicates that assumptions of "normal" sleep practices and disorders need addressing. To date, the representation of sleep with regard to aging has yet to be explored in this way. Therefore, the present research sought to answer the question: how does New Zealand's mainstream online news media construct sleep and aging?

Methodology

Critical Discourse Analysis

This research examined sleep messaging in the media using Fairclough's approach to Critical Discourse Analysis (CDA). Discourse refers to the broad systems of meaning that shape the use of language in a particular social context. The media has a predominant role in the scope of different discourses and the capacity to accentuate and proliferate dominant ideology (Fairclough, 2001). This includes shaping beliefs and expectations around time use, health, and life stage. Therefore, an analysis focusing on discourse as a product of social interaction within context is essential in understanding the power relations and dominance of media messaging regarding sleep and aging. A CDA framework focuses on three dimensions: text, discursive practice, and social practice (described in Table 1). Fairclough views discourse as mediating the link between text and social or cultural practice. A text is shaped by its production process, and interpretation operates by the "cues" in

Table 1. Summary of the Three Dimensions of Critical Discourse Analysis (Fairclough, 2001; Janks, 1997) and Interpretation into Analysis of the Sleep and Aging Content Within Stuff Articles

Dimension	Process	Details	Analytical approach
1. Text	Description	Defining a "text," which can be written or spoken language which is produced within a discursive event. Texts are described including language and the grammatical formulations.	Information about each article was entered into an Excel spreadsheet, for example, including title, key words, section of stuff, intended audience. Attention to how content and form is used to convey meaning.
2. Discursive practice	Interpretation	Interpreting how texts' production, distribution, and consumption came into being. Draw attention to how repertoires, genres, and discourses are drawn upon and woven into a text. Dimensions 1 and 2 support one another as evidence for meaning making. This allows for the identification of discourses in a text, the tensions between discourses, and to demonstrate ideological forces at work.	All contents related to sleep and older people in the articles were coded in a qualitative data analysis software programme (NVivo12). Codes related to the experience of sleep among older people, diseases experienced as people age and links to sleep, and remedies suggested to address problematic sleep in older age. Annotations were added to explain the context surrounding the section of text coded and notes added as memos.
3. Social practice	Explanation	An examination of the socio-historical conditions of the production and interpretation of discourse. Text and language use is considered a social process and a text is shaped by its surrounding social structures. This dimension aids in the identification of predominant discourse and their reproduction or reconstruction as representing and explaining social practices.	Following coding, repetitive patterns in the ways of describing sleep and aging were noted. This process considered the relationships between each dimension to understand how text are connected with specific forms of social and discursive practice, leading to the identification of key discourses in the set of texts.

the text. However, both the production and interpretation are shaped by and help shape social and cultural contexts (Fairclough, 1992). The three dimensions, therefore, intersect, and their interconnection allows for the identification of tensions in the constructs of sleep and aging that can be described, interpreted, and explained. As Fairclough's dimensions are mutually explanatory, this approach allows for multiple points of entry with all dimensions being considered rather than isolating each one (Janks, 1997). The CDA approach can, therefore, reveal the social and cultural contexts and how a change in either of these practices or processes influences one another.

Media Data: Stuff.co.nz

Online content is the primary information source in society (Mitchelstein & Boczkowski, 2009). Stuff.co.nz ("Stuff") was the platform chosen for this analysis as it provides free online content and consequently has a wide readership (reaching 2.3 million New Zealanders, approximately 45% of the population, monthly, Stuff Limited, 2020), and has broad accessibility compared to print articles or subscription-based platforms. Stuff's readership is evenly split between men (45%) and women (55%), with those aged 55 and over making up 31% of the audience (Stuff Limited & Fairfax Media, 2017). As well as news content, Stuff has sections on lifestyle, travel, beauty, health, and well-being.

Search Procedure and Analytical Approach

For the present analysis, online articles published on Stuff within a recent 3-year span (December 2018–December 2021) were considered. Key search terms were informed by previous literature on sleep and aging. The terms included '"aging," '"older," '"elderly," and '"dementia," and their variants. Searches were conducted by combining each age descriptor with the word '"sleep"' (e.g., sleep AND aging). Duplicate articles were excluded as well as those that did not apply both words or discuss sleep-related content in the context of aging or older people. After the screening, a total of 38 articles remained (see Figure 1). Table 2 provides descriptive information on the articles. Most (74%) were published under Stuff's lifestyle section and others under either national news, travel, or world news sections (see Table 2). Table 1 provides a summary of the three dimensions of CDA and their application to the present study with regard to analysis of relevant texts within the articles. All sleep-relevant excerpts from the news articles were coded using NVivo12 software. Content was then organized under common thematic patterns of sleep and: "general aging" (27 references across 9 articles), "disease" (27 references across 8 articles), "appearance or beauty" (8 references across 5 articles), "approaches to treatment and the medicalization of sleep" (18 references across 8 articles), as well as "other" miscellaneous content (8 references across 6 articles). This provided a foundation for analyzing the texts as sets to identify discourses that were used within and across themes. This resulted in the classification of three dominant discourses that were used to describe the intersection of sleep and aging and an analysis of how their production represents and explains broader social practices.

Results

Three key discourses were identified in the set of texts: "Aging inevitably changes sleep," "sleep as a magic bullet," and "it's just [complicated] sleep." Each discourse is described and discussed with exemplar extracts from the media data below.

Aging inevitably Changes Sleep

A predominant discourse in the identified texts was that poorer sleep is an inevitable part of aging. Sleep in later life was presented as of both poor quality and reduced quantity. One text commented that older people commonly struggle to get enough sleep, while another mentioned that even getting the right amount of sleep would be insufficient as sleep

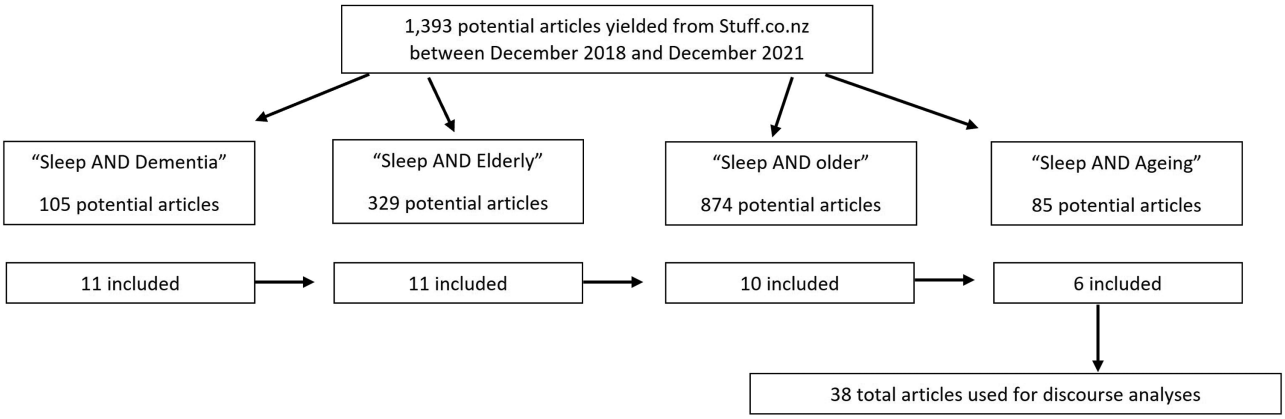

Figure 1. Search results of Stuff.co.nz articles by keywords.

quality declines with age. The following extract challenges the belief that the need for sleep declines as people age, but difficulty achieving good sleep is linked to inevitable change with aging:

> *It's a myth that we need less sleep as we age, according to Leschziner."Older people probably need just as much as younger adults, but they're less good at achieving it," he says.* (Article 14)

Situated within an article on how to improve sleep at any age, the section concerning older people focuses primarily on managing the inevitable decline in sleep with aging. One age-related sleep change is framed as a myth by aligning the needs of older people as the same as younger people. The issue is replaced with a collective "they" of older people as deficient in maintaining sleep relative to younger people.

General physiological decline with age was also linked to declining sleep quality and quantity, with Article 14 going on to describe how aging brains become "less good at maintaining sleep." The inevitability of sleep changes with age is reinforced by reports on rates of sleep deficit and disorder among older people:

> *In New Zealand's National Health Survey 2018-19, it was revealed that 31 per cent of Kiwis [New Zealanders] over the age of 15 fail to get the recommended amount of sleep each night. That number rose in older people.* (Article 16)

By phrasing sleep deficit in terms of a failure to get recommended amounts, sleep is framed as a personal responsibility for all age groups, despite the construction of sleep decline with aging as inevitable and as linked to physiological changes. Sleep disruption was also associated with experiences of chronic pain, which is also described as an age-related problem that is common knowledge: "Unsurprisingly, the figures show prevalence [of chronic pain] does increase with age" (Article 24). Together this suggests that sleep worsens with age alongside the general experience of physical decline, and frames these as changes that readers should expect and consider inevitable.

Sleep changes with aging were also linked to life course transitions. For example, the loss of structured routines was linked to aging through withdrawal from paid employment:

> *Wilson says retirement can also trigger problems. "For a lot of people, their job gives them their clock. Things you might have done when you were working, whether it was watching the news right before bed, drinking alcohol or a coffee late in the day, might start to be more of a problem because you're no longer as physically and mentally tired. Getting into the habit of late nights and lie-ins can confuse the body clock."* (Article 14)

Retirement was linked to a lack of routine through the assumption that it will mean reduced physical and mental activities, late nights, and sleeping in. This excerpt characterizes retirement as a time of inactivity and indolence and moralizes sleep changes by linking them to poor habits which create physiological confusion. The use of "you" throughout places retirees as the intended audience of these messages, creating a distinct group responsible for replacing work routines with healthy retirement habits.

Sleep as a Magic Bullet

Sleep was regularly discussed in the media as a remedy for addressing a range of aging-related issues and illnesses. The texts use quotes and opinions from health professionals and lay people who promote good sleep as a tool for healthy aging. The text below is from an article described as science-based, with the headline "Scientists reveal the secrets to a restorative sleep." However, it ironically asserts that the key to living longer and healthier lives is simple and accessible to most people, as long as good sleep quality and quantity are attained. Rather than secrets for sleep, sleep was framed to the audience as the cure for a list of health conditions and neglecting sleep was linked to poor quality of life:

> *If someone promised you a revolutionary new treatment that made you live longer, boosted your memory and creativity, kept you slim and made you better looking, would you be interested? And wait, it also helps guard you from cancer and dementia, lowers your risk of heart attacks, stroke and diabetes, and keeps away colds and the flu. Sound good? You'll even feel happier, less depressed and less anxious. Of course you'd take it. It's just sleep. Eight hours a night of solid, uninterrupted sleep.* (Article 22)

Here, good sleep was promoted as a lifestyle factor available to all that protects against signs of aging, disease, and illness,

with the common media marketing phrase of "And wait, it also …." This was used to frame the benefits as numerous and pervasive whilst also illustrating the solution as simple. This extract hinges on the simplicity of sleep "It's just sleep," and the achievement of uninterrupted sleep as something that people can "take" like a form of medication or supplement. Discussion of sleep as a remedy was often followed by comments on good sleep as accessible to all by improving sleep habits. The notion of sleep as the answer to healthy aging is further illustrated by media connections between geographical areas where people live longer and healthier lives, for example, "the famed Blue Zones" (Article 7) where rest and sleep are part of what distinguish the lives of older people in these communities. Sleep is described as key to addressing the health problems of modern society in the texts, for example: "Sleep is something we often underestimate as a protection from many forms of ill health" (Article 9). This frames sleep deficit as due to a lack of knowledge and frames the problem as needing to increase recognition around the pervasive benefits of sleep to prevent or delay age-related decline.

The counterpoint to the "sleep as a magic bullet" discourse was the presentation of the dangers of poor sleep quality or insufficient sleep; difficulties maintaining sleep were linked to illness and disease. The increased risk of developing Alzheimer's disease and dementia was often chosen to illustrate this connection:

> *A new study supports a suspicion long held by researchers and doctors—the amount of sleep we get could impact our likelihood of developing dementia in old age.* (Article 18)
> and
> *Routinely sleeping less than six or seven hours smashes our immune systems, more than doubling the risk of cancer, he says. It is a key lifestyle factor in determining whether we develop Alzheimer's disease.* (Article 22)

Against the positive framing of sleep as improving all aspects of health and well-being, the negative framing of the dangers of poor sleep disproportionately focuses on dementia risk. The second extract uses both epidemiological terminology on risk ratios alongside lay assessment of low sleep as "smashing" the immune system, making the argument both scientifically convincing and compelling. The use of "determining" to describe the influence of sleep on Alzheimer's risk strengthens the causal link whilst the use of "we" focuses on this as a risk we all share. Dementia functions here as an example of a feared condition in later life which works to emphasize the significance of sleep to help overcome age-related disease.

"It's Just [Complicated] Sleep"

The third discourse used in the media articles was of sleep as a complex issue with simple solutions. In the discourse of "it's just [complicated] sleep," the physiological need for sleep was positioned against the societal expectation of constant progress and achievement and this was presented as a difficult negotiation, especially for women:

> *Society's "culture of busyness" didn't help. "There's this belief that we can sleep when we're dead, that we all have to live big, busy lives, especially women who are more likely to suffer from insomnia than men, probably because society expects us to be amazing on so many levels. But the reality is around one-third of Kiwi [New Zealander] women aged between 30 and 60 are trying to be amazing on not enough sleep.* (Article 27)

Sleep problems were presented as both ubiquitous and beyond the power of individual resistance ("society's culture (…) to live big, busy lives") and gendered due to women experiencing greater pressure "to be amazing on so many levels." High expectations for women create a complex issue of attempting to balance living up to these expectations whilst also getting enough sleep. The imbalance of these expectations leads to a circular argument: women experience insomnia because of societal expectations and lack of sleep impedes their ability to be amazing. This article was published in the lifestyle wellbeing section and focuses on how one high achieving woman overcame insomnia through cognitive training and succeeded in her personal and professional life. None of this argument is used to question or undermine the gender disparity that fuels insomnia or to question the desire to be "amazing" that undermines sleep health. This was further linked to physiological changes at menopause:

> *Poor sleep and tiredness – From talking to my female patients, this is often one of the hardest things to cope with; sleep patterns can become disrupted either because of flushes and sweats, or for no apparent reason at all.* (Article 19)

Published in Stuff's "Lifestyle—Teach me" section, this extract presents sleep for older women as a complicated and unpredictable issue. Following the presentation of the complex issues of sleep, solutions were commonly offered immediately. These solutions usually revolved around changing individual sleep habits or giving simple generic tips to improve sleep routines. Despite the complexity of getting good sleep, texts often simplified the process of achieving good sleep by stating it as a factor that just needs management. The following text is from an article focusing on the potential link between Alzheimer's disease and menopause, which acknowledges the complexity of a deteriorating brain and problems pinpointing its cause. Regardless of this complexity, the management of lifestyle factors such as sleep is promoted:

> *"There's a complex interaction of lifestyle, education, stress and other factors, it's very difficult to pick out just one as a single cause of brain deterioration." She does, however, agree with Martins on the importance of managing lifestyle factors such as diet, exercise and sleep.* (Article 9)

This discursive construction relies on acknowledging complexity and context for plausibility, it would be implausible to propose a singular cause of brain health. Crucially, this list of indefinite and open causes "lifestyle, education, stress and other factors" and their intersection is used as the basis for messaging on managing personal factors including sleep habits which do not address the listed factors. Proposing good sleep habits to combat the complex issues older people may have with sleep regularly included recognition that many proposed solutions are ineffective:

> *Those who have difficulty sleeping will know the symptoms, the endless remedies, from warm baths to bananas to shakti mats, from nightcaps to giving up booze and coffee, to demanding melatonin or sleeping pills from their doctor.* (Article 22)

Listing a variety of sleep remedies from pleasant activities like warm baths to confronting suggestions of demanding medications from the doctor and positioning this as "endless" functions as an extreme case formulation. It emphasizes the limited effectiveness of all health promotion messaging for sleep but, simultaneously, the infinite options on offer. People who have difficulty sleeping are positioned as the experts who know from experience that the simple tips commonly prescribed do not work. However, recognition of ineffectiveness is counter-intuitively the basis for further health promotion messaging:

> *Although it is important to try to stick to a good routine and encourage your body to sleep as much as you can, this won't always work. The longer it goes on, the more likely the chronic tiredness is to impact on your mood and ability to function.* (Article 19)

Maintenance of a good sleep routine is promoted alongside the limitations of this advice, and the final message is the danger of tiredness to functioning. This discursive construction relies on acknowledging the complexity of sleep disruption to establish credibility and avoid counterarguments from the intended audience that sleep is not that simple. By acknowledging the complexity of sleep and the potential futility of efforts to improve it, whilst also advocating sleep hygiene as "important to try," older people are implored to manage their sleep health. Together, the audience of these complex messages is left in the invidious position of both pursuing simple sleep practices to prevent age-related sleep decline, whilst simultaneously being told that degradation of sleep is inevitable.

Discussion

This manuscript represents novel considerations of how sleep and aging is represented in mainstream online media. Three key discourses have identified that work together. Sleep is presented as inevitably declining with aging through physiological changes and life transitions; sleep is given as a cure for ill health and processes of aging; and the complexity of sleep is presented as the foundation of simple tips for improving sleep with age. These discourses appear to provide mixed messages whilst working together to attribute agency to older people in managing their sleep and health. Here negative changes to sleep are suggested as inevitable and to be expected with aging, while also presenting solutions as individualized, straightforward, and accessible. These kinds of discourses operate across health promotion messaging for older people and as well as for younger audiences to prepare for later life (Swacha, 2017). Swacha (2017) refers to this coupling of a serious problem with a simple solution in health promotion materials as a "good news" trope; that solutions to health challenges are available and overcoming risks is easy. This presents a complex discursive environment to navigate. Simple tips, together with acknowledging complexity and inevitable decline, function as prescriptive advice which may not ameliorate the complex physiological and societal foundations for sleep change in later life.

Sleep is a behavioral outcome of biological, behavioral, and social factors which changes over the life course (Assefa et al., 2015; Grandner, 2019). Considering media representations of sleep provides a useful lens for understanding discourses of aging. The media has a dominant role in promoting societal ideologies and frequently supports societal stereotypes (Edström, 2018; Vasil & Wass, 1993). Older people are often framed as innately vulnerable (instead of situationally vulnerable), and aging is associated with declining health (Makita et al., 2021). Older people were described as experiencing an inevitable decline in their sleep and retirement was presented as a problem for sleep health, rather than an opportunity to introduce healthy habits. Although empirical research indicates that changes to the structure of sleep and an increased incidence of sleep problems are associated with advancing age (Ancoli-Israel et al., 2008; Roepke & Ancoli-Israel, 2010), the proliferation of such messages through the media may promote problematic beliefs (Robbins, Grandner, et al., 2019). This includes both the normalization regarding sleep change identified in previous interview studies among older New Zealanders (Crestani et al., 2022) and the hopelessness experienced by families living with dementia (Gibson et al., 2014, 2023). As people age, negative stereotyping and stigmatizing of older people can affect beliefs about their abilities and capacities (Breheny & Stephens, 2019; Edström, 2018; Vasil & Wass, 1993). When decline is viewed as a personal responsibility to manage or an inevitable process, this has implications for help-seeking and attaining the necessary support to improve sleep when it would be beneficial.

Lifestyle practices such as sleep are often targeted as a risk factor to prevent ill-health, or disease in later life (Armstrong, 1995; Meadows et al., 2018). In the present analyses, dementia was commonly used to illustrate a negative outcome associated with poor or reduced sleep. This adds to previous research illustrating the negative connotations, discourses, and stigmas associated with Alzheimer's and other dementia-related diseases which are widespread in mainstream media and caregiver narratives (Behuniak, 2011; Low & Purwaningrum, 2020; Peel, 2014). Here sleep loss was identified as a threat for dementia later in life, however minimal useful guidance was provided on realistic alleviation. As such, the link between sleep disturbance and dementia is used to scare older people to prioritize sleep, further entrenching dementia as a symbolic fear in later life.

The present study indicates how sleep is presented as both a complex and a simple solution to maintaining health and well-being. Attaining good sleep was labeled an individual responsibility across the lifespan. Others have also noted how people appear to bear all accountability for sleep, and how moral judgments are made for failing to maintain good sleep and avoid the adverse consequences of sleep deficit (Kroll-Smith, 2003; Zarhin, 2021). Such an onus being placed on individuals is an example of neoliberal approaches to health management (Allain & Marshall, 2017). The maintenance of sleep is emphasized as having implications for not only individual health but also waking productivity including social and economic engagement.

Emphasizing individual responsibility for good sleep and identifying poor sleep as a health risk has resulted in the practice of sleep being objectified (Hislop & Arber, 2003; Williams & Boden, 2004). Sleep is increasingly characterized as a commodity and recommendations are made around "doing" sleep to optimize health and functionality which has resulted in a growing market for products and remedies with the promise to track, induce and/or or improve sleep (Barbee et al., 2018; Meadows et al., 2020; Williams & Boden, 2004; Williams et al., 2021). However, sleep trackers have limited reliability due to their reliance on accelerometry to infer sleep and wake and reduced accuracy with aging and disease (Wei &Boger 2021). Furthermore, their data seldom align with subjective sleep status (Landry et al., 2015). Therefore, tracking and self-monitoring of sleep may bring limited benefits for improving sleep experiences. Tracking offers the user a promise of improved functionality but may instead induce a "tyranny of healthy activities and lifestyle expectations" (Katz & Marshall, 2018, p. 64). Although older subjects are presented as responsible for their bodies and performance (Katz, 2000), behaviors like sleep tracking need consideration alongside individual social and contextual circumstances which can all act as barriers or facilitators to achieving good sleep.

The media often described sleep as a resource that requires development alongside other aspects of the self and the body. This was compounded by many of the present articles being published in the "teach me" section of Stuff, which suggests that good sleep can be taught. Individual and social circumstances are often excluded from media messaging (Robbins, Grandner et al., 2019). This is apparent in the present study, with simple solutions proposed to overcome the complex sleep problems experienced by older people. Good sleep is often shaped by social and economic status (Taylor, 1993); many people struggle to create conditions for supporting the dimensions of sleep health (Hale et al., 2020). The complexities of sleep, particularly insomnia-related symptoms, appear to be especially apparent among women (Williams et al., 2008a). Previous works highlight how women are led to believe that their sleep will deteriorate with age due to a range of physical, psychological, and social factors whilst higher social expectations also impinge on their ability to prioritize good sleep (Hislop & Arber, 2003). In this study, women were encouraged to use sleep to meet unrealistic societal expectations. Such discourses gender sleep problems and solutions without examining the norms and expectations that underpin the conditions under which women sleep. For example, older women have been identified as more likely to undertake caregiving duties than older men (Alpass et al., 2013), roles that impinge on the quality and quantity of sleep (Gibson et al., 2023).

People are more likely to have access to and be influenced by information from popular media sources rather than scientific research. However, they also vary in how they accept, alter, or resist the media they access (Breheny & Severinsen, 2018; Dew et al., 2016). Responses to media messages may not be uniform, they depend upon how audiences see themselves positioned by these media representations. Older people in good health benefit from their position as having earned admiration for their good health (Breheny & Stephens, 2019). An important next step would be to understand audience consumption, interpretation, and response to media discourses. This might include the mismatch between marketing sleep as a modifiable behavior without necessarily improving it. Such analyses could be used to inform more appropriate and reflective sleep content in the media.

Abroms and Maibach (2008) note that media communication to improve public health tends to focus on individual behavior change. They note a missed opportunity to improve public health through mass media communications which aim to directly alter the community norms and public policies that underpin health (Abroms & Maibach, 2008). Hale et al. (2020) include the media as important in promoting a culture of sleep health which goes beyond individual health promotion and acknowledges health inequities. Key to such mass media advocacy is identifying the default frames in media health messaging, such as individual responsibility for health outcomes, and advocating for health research messaging that provides alternative frames that highlight inequities (Dorfman & Krasnow, 2014). Such advocacy efforts should not be overlooked because they represent ambitious goals; they have the potential to create sustainable improvement in health through structural change.

Limitations

Older people are not typically viewed as newsworthy (Edström, 2018) and this pattern is held in the present dataset with most of the articles selected including only partial material on sleep and aging rather than whole articles relevant to older people. Given this, the media data collected does not provide a comprehensive account of older people's sleep. However, it does have the advantage of wide coverage across a range of articles and the presentation of different constructions of older people alongside sleep content for broader audiences. Much of the sleep-related content in this analysis was published in lifestyle sections explicitly designed for health promotion. This kind of journalism is niche and is commonly the site of simplistic messaging from more nuanced research findings or interventions on health and wellbeing (Dew et al., 2016; Lupton & Chapman, 1995). Alternative media platforms or longer-format magazines may reproduce different discursive constructions of sleep and aging and be applicable to broader audiences, warranting consideration in future research.

The relationship between ethnicity and sleep health is increasingly recognized. New Zealand research indicates that Māori and Pacific peoples are significantly more likely to have restricted or disordered sleep compared with non-Māori/non-Pacific people. This has been associated with inequities in health as well as living and sleeping environments (Muller et al., 2019; Paine & Gander, 2016). Māori research participants may not so readily pathologize sleep disruptions compared to Western views of sleep disturbance. Sleeplessness is also viewed as providing an opportunity for cultural or spiritual connection (Crestani et al., 2022). Future research could include alternative sources (e.g., radio, television, and arts) created by and for other audiences including Indigenous peoples (Rankine et al., 2014). Broadening the search strategy to include such media would enable more detailed understandings of sleep, aging, and culture which have potential to represent and improve messaging for all.

Table 2. Summary and Reference Data for Articles Included in the Present Analyses

ID	Title	Date (D/M/Y)	Published	Article code
1	What happens as we die?	20/09/2019	• Stuff > lifestyle	115136302
2	Top chef Des Harris on leaving fine dining and finding balance	30/12/2018	• Stuff > lifestyle	108771498
3	What's wrong with looking my age?	2/28/2019	• Stuff > lifestyle > beauty	110932042
4	Are collagen supplements a waste of money?	5/02/2000	• Stuff > lifestyle > beauty	119185903
5	"Beauty" supplements: Drinking to better skin or a waste of money?	17/01/2019	• Stuff > lifestyle > beauty	109986433
6	Silk pillowcases claim to improve sleep, skin, hair. But do they work?	17/02/2021	• Stuff > lifestyle > beauty	300232892
7	The Better Base: using a plant-based diet to combat disease	1/06/2019	• Stuff > lifestyle > food and wine	113143417
8	Sleep better with an aromatic, soothing herbal pillow	1/04/2020	• Stuff > lifestyle > garden	119117589
9	Possible "link" between menopause and Alzheimer's is scaring women, including me	7/06/2019	• Stuff > lifestyle > life	113301561
10	An open letter to Jacinda Ardern as we approach our 40th birthday	24/07/2020	• Stuff > lifestyle > life	122235345
11	Retirement: How to embrace a lifestyle with its own rhythms	6/08/2020	• Stuff > lifestyle > retirement	300059659
12	Retirement: I'm less than 20 years away from retiring and I'm terrified	7/08/2020	• Stuff > lifestyle > retirement	122333261
13	Why skin sags in your 40s (and what to do about it)	25/09/2019	• Stuff > lifestyle > skincare	116032356
14	How to get a good night's sleep - whatever your age	23/07/2019	• Stuff > lifestyle > teach me	114306425
15	What you need to know about Parkinson's Disease	3/12/2019	• Stuff > lifestyle > teach me	117856275
16	Being Human: How to get a better night's sleep	13/02/2020	• Stuff > lifestyle > teach me	118531928
17	Is chamomile really good for you?	23/02/2020	• Stuff > lifestyle > teach me	119608219
18	Dementia and poor sleep could be linked, new study finds	27/04/2021	• Stuff > lifestyle > teach me	300284193
19	How to tell if you are going through menopause	7/01/2020	• Stuff > lifestyle > teach me	118570188
20	Why is immortality once again big business and what does that say about us?	31/08/2019	• Stuff > lifestyle > teach me	115174083
21	Say goodbye to dark under-eye bags. A definitive guide	27/01/2021	• Stuff > lifestyle > teach me	300208357
22	Scientists reveal the secrets to a restorative sleep	23/01/2019	• Stuff > lifestyle > well and good	110106372
23	How to spot the signs of dementia in a loved one	14/07/2019	• Stuff > lifestyle > well and good	114226818
24	The pain question: Why does it hurt to get older? Why does it hurt at all?	7/11/2020	• Stuff > lifestyle > well and good	123235373
25	Do we really need 8 hr sleep, 5+ fruit and veg, and a litre of water?	30/07/2019	• Stuff > lifestyle > well and good	114625792
26	Menopause made easier over a martini	15/10/2020	• Stuff > lifestyle > well and good	300131657
27	Sleep easy: How one Kiwi finally conquered her chronic insomnia	4/05/2021	• Stuff > lifestyle > well and good	125021332
28	When push comes to love - parenting your parents	6/11/2021	• Stuff > lifestyle > wellbeing	300446094
29	Late wife's memory spurs on 78-year-old Mt Taranaki climber again and again	12/03/2019	• Stuff > national	111180660
30	Sexual predator with attraction to "deviant sex" jailed over drugging attempt	1/02/2019	• Stuff > national	110251419
31	Calls to stop using anti-psychotics as "chemical restraints" for people with dementia	15/12/2019	• Stuff > national > health	117524333
32	Some rest homes are rejecting anti-psychotics in favor of person-centred care	16/12/2019	• Stuff > national > health	117927587
33	Rest home evicted a 90-year-old after her family raised concerns about her care	1/05/2020	• Stuff > national > health	119661752
34	The elixir of life: How exercise slows down the aging process	11/04/2021	• Stuff > national > health	300268756
35	Dementia carers "haven't slept in years," suffering burnout	23/04/2021	• Stuff > national > health	124880854
36	"This is the last straw"—rest home residents given six weeks to move out	19/02/2019	• Stuff > the press > news	110666300
37	Travel tips: How to make economy feel like first class	7/01/2020	• Stuff > travel	118632756
38	The extraordinary battle over an aging Hollywood titan's care	10/05/2019	• Stuff > world > Americas	112627870

Notes: ID = identification number; D/M/Y = day/month/year

Conclusion

This research demonstrates the complexity of media messaging regarding sleep and the fraught options it offers people as they age. Good sleep with aging is presented as a reasonable achievement to strive for while also inevitably declining. This messaging mirrors the health messaging that older people receive in general: that they are both responsible for resisting aging and at the same time falling into inevitable decline. The repeated messages of simple fixes alongside acknowledged complexity produce powerful imperatives for older people to take control of their lives and sleep habits to avoid increased risks of illness and disease. Media articles do acknowledge complexity and this could be the starting point for more nuanced sleep messaging that acknowledges the context in which such practices are embedded. Acknowledging the social context of sleep with aging may better relieve older people of such imperatives to manage health and provide better support for rest and well-being in later life.

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Conflict of Interest

None declared.

Data Availability

This work is part of a larger research project: "Sensationalising sleep: Discourses and practices of sleep in Aotearoa" which has subsequently been funded by the New Zealand Royal Society. All data used for the present analyses are available from Stuff.co.nz. References concerning critical discourse analysis methods are provided.

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